Name: ______________________

Worksheet

Comparing summaries of data

Task A: Using a summary statistic to compare datasets

1) Lucas is checking his progress in maths from year 7 to year 8. He calculates the median and range of his scores in his weekly quizzes.

Match the summary statistic to its correct interpretation.

	year 7	year 8
median	16	17
range	7	5

a His middle score in year 8 was higher than in year 7.

b His middle score in year 7 was higher than in year 8.

c His scores were more varied in year 8 than in year 7.

d His scores were more varied in year 7 than in year 8. *range*

e His middle score was the same in both years.

2) Lucas is checking his progress in English from year 7 to year 8. He wants to calculate the mean score of all his homework.

year 7	46	52	51	65	42	59
year 8	37	42	51	71	69	63

a) Find the mean of both his year 7 and year 8 English homework scores.

b) Use the values of these means to write a sentence explaining whether his English homework scores improved over the two years.

Task B: Using multiple statistics to improve comparisons

1) The table of summary statistics shows the number of goals scored by two footballers each match.

Which combination of these statements creates the correct comparison of these results?

	Ella	Wayne
mean	2.9	2.4
median	2	3
mode	3	2
range	3	4

a Ella had less variance in the number of goals scored than Wayne

b Wayne had less variance in the number of goals scored than Ella

c And Wayne's most frequent number of goals was higher

d And Ella's most frequent number of goals was higher

e Therefore, Ella scored more goals on average than Wayne.

f Therefore, Wayne scored more goals on average than Ella.

2) The table of summary statistics shows the number of days of air frost each day in December over the last 10 years.

Complete the sentences with either “Oxford”, “Stornoway” or a correct number to accurately compare the two locations.

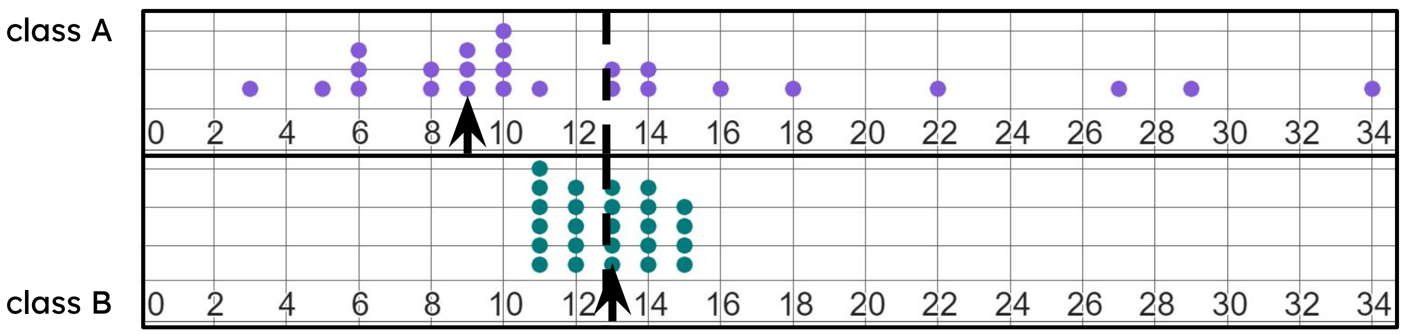
	Oxford	**Stornoway**
mean	5.8	3.1
median	6	2
mode	bimodal at 0 and 6	bimodal at 1 and 2
range	12	7

The mean number of days of air frost in __________ is lower by 3.7 days.

The range of 12 in ___________ shows a lot more variability in the number of days of frost each year.

In conclusion, there was more frost in __________ , supported by the fact the middle value of __________ is higher by _____ days.

3) The dotplot shows the number of minutes each student in two classes takes to walk to school, with the mean shown by a dotted line and the median shown by an arrow.



a) Complete the statement below about the summary statistics for class A and B.

The average, typical value for class A and class B are ___________________ , with a value of approximately _________ .

b) Complete the statements below about the summary statistics for class A and B.

However, class B had a much higher __________ of ______ compared to ______ .

Class B also had a slightly higher mode of ______ compared to ______ .

c) Complete the statement below about the summary statistics for class A and B.

Class ___ is much more varied, with a _______ of ______ compared to ______ .

d) Complete the concluding statement below about the summary statistics for class A and B.

In conclusion, I think class ___ takes longer on average to walk to school, as their ________ and ________ are larger, and there is also less variation in these longer central times.

4) The two lists of data show the number of days of air frost each January.

Complete the table of summary statistics and interpretations, and then write a conclusion to the question: "which location had a greater average number of days of air frost each January?"

Tiree (Scotland)	0	0	0	1	1	2	3	4	11			
Valley (Wales)	0	1	1	1	2	4	4	5	6	8	10	11

	Tiree	Valley	interpretation
mean		4.42	
median			The middle number of days of air frost in Valley is higher than in Tiree
mode			
range	11	11	